Chaos-Informed NGRC–TCN Encoder–Decoder for Multi-Step Waveform Forecasting

Validation on the Lorenz System and Application to ECG

NU Research Internship 2026 — Time Series Forecasting Group

Ibrahim Hanafy
Zewail City of Science and Technology

August 2026

Abstract

This report documents the design, failure, diagnosis, and successful validation of a chaos-informed forecasting architecture. The model applies Phase Space Reconstruction (PSR) and a Next-Generation Reservoir Computing (NGRC) polynomial feature map as a zero-parameter front-end to a Temporal Convolutional Network (TCN) encoder–decoder. An initial application to ECG waveform forecasting failed, returning $R^2 = -0.082$. Rather than tuning blindly, the same architecture was validated on the Lorenz system, where the largest Lyapunov exponent is known exactly and forecast horizons can be reported in Lyapunov time. After correcting an input-length pathology in the decoder, the model reached $R^2 = 0.995$ at 0.23 Lyapunov times and $R^2 = 0.956$ at one full Lyapunov time, outperforming a raw-input TCN control (0.738), a recursive NGRC ridge model (0.157), a linear autoregressive baseline (0.246), and persistence (-0.434). Validation includes a shuffled-target leakage control, attractor reconstruction, and Lyapunov exponent recovery. The ECG failure is therefore attributable to properties of the ECG task rather than to an implementation fault, and is shown to arise from stochastic beat-timing variability interacting with a squared-error objective.

Contents

1 Introduction **3**
1.1 Contributions 3

2 Literature Review **3**
2.1 Chaotic Characterisation of ECG and Phase Space Reconstruction 3
2.2 Reservoir Computing and NGRC 4
2.3 Temporal Convolutional Networks 4
2.4 Identified Gap 4

3 Method **5**
3.1 Architecture 5
3.2 Evaluation Protocol 5

4 Experimental Trials **5**
4.1 Trial 1 — Initial ECG Forecasting (Failure) 5
4.2 Trial 2 — Lorenz Validation 5
4.3 Trial 3 — Input-Length Ablation 6
4.4 Trial 4 — Corrected Lorenz Sweep and Full Ablation 6
4.5 Validation Checks 7

4.6 Comparison at FGNet Horizons 7
4.7 ECG Baselines and the Baseline-Wander Effect 8

5 Findings **8**
5.1 Why ECG Forecasting Fails at Multi-Second Horizons 8
5.2 Chaos Does Not Explain the ECG Failure 8

6 Limitations and Next Steps **9**

1 Introduction

Electrocardiogram (ECG) signals encode cardiac electrical activity as quasi-periodic waveforms. Research overwhelmingly targets classification; forecasting raw future waveform amplitude remains comparatively underexplored. ECG has been widely described as exhibiting chaotic dynamics, which motivates applying tools from nonlinear dynamics — delay embedding, Lyapunov exponents, polynomial feature maps — to the forecasting problem.

This report is organised around an experimental narrative rather than a single result. Section 2 reviews the relevant literature. Section 3 states the architecture. Section 4 documents every trial in chronological order, including the failures, because the diagnostic path is the main contribution. Section 5 states the findings and Section 6 the limitations.

1.1 Contributions

1. An NGRC polynomial front-end feeding a MIMO TCN encoder–decoder, evaluated against a matched raw-input control rather than against unrelated baselines.

2. Evaluation across a sweep of forecast horizons expressed in Lyapunov time, rather than at a single arbitrary horizon.

3. Identification of an input-length pathology in interpolation-based decoders: reducing input length from 540 to 64 samples raised R^2 at one Lyapunov time from 0.34 to 0.96.

4. A mechanistic account of why ECG waveform forecasting fails at multi-second horizons, supported by autocorrelation evidence rather than assertion.

5. An observation that ECG forecasting results reported without high-pass filtering are inflated by baseline wander.

2 Literature Review

2.1 Chaotic Characterisation of ECG and Phase Space Reconstruction

ECG signals arise from nonlinear interactions in the cardiac conduction system, and have been characterised as chaotic through Lyapunov exponents, correlation dimension, and attractor geometry [6]. Takens' delay embedding theorem provides the formal basis for reconstructing an attractor from a scalar observable:

$$\mathbf{v}(t) = \big[x(t),\ x(t-\tau),\ \ldots,\ x(t-(d-1)\tau)\big], \tag{1}$$

where the delay τ is conventionally selected from the first minimum of Average Mutual Information (AMI) and the embedding dimension d from the False Nearest Neighbours (FNN) criterion.

Phien et al. [5] applied AMI and FNN before feeding embedded vectors to LSTM autoencoders, and compared five multi-step strategies (recursive, direct, MIMO, DirRec, DirMO), concluding that MIMO performed best and recursive worst due to error accumulation. This directly motivates the MIMO design adopted here. Khoshroo and Shekofteh [4] converted PSR embeddings into recurrence plots and combined them with a temporal branch, reporting a 36% RMSE reduction on MIT-BIH. It is worth noting that their pipeline applies no filtering to the raw ECG before constructing recurrence plots; since recurrence plots depend on an ε distance threshold, baseline wander directly distorts the recurrence structure.

2.2 Reservoir Computing and NGRC

Reservoir computing fixes a high-dimensional nonlinear dynamical system and trains only a linear readout. Gauthier et al. [3] removed the random reservoir entirely, constructing features directly from delayed observations and their polynomial cross-terms. On Lorenz-63 this achieved NRMSE $\approx 1.06 \times 10^{-4}$ with training in under 10 ms. Critically, NGRC operates recursively: a one-step map is fitted, then the prediction is fed back so the system runs autonomously.

Shahi et al. [7] formalised NGRC as a Nonlinear Vector Autoregression and compared LSTM, GRU, ESN and NVAR on chaotic systems including zebrafish cardiac voltage recordings, finding gated RNNs to underperform on chaotic forecasting relative to reservoir approaches.

An important limitation for the univariate case is documented in the partial-observation NGRC literature: when only one state variable is observed and time-delay coordinates are used, degree-2 monomials are insufficient, and higher-degree bases (e.g. Chebyshev polynomials of degree 9) are required. This predicts that a degree-2 univariate NGRC will underperform, which is confirmed experimentally in Section 4.

Wang et al. [8] applied a minimalistic memristive reservoir to ECG classification, reporting 98.3% accuracy — though on a test set of only 60 beats, with confidence intervals overlapping the baseline it claims to exceed.

2.3 Temporal Convolutional Networks

TCNs offer parallelisable training, stable gradients, and exponentially growing receptive fields via dilated causal convolutions. Dudukcu et al. [1] benchmarked hybrid TCN–LSTM and TCN–GRU models against twelve alternatives on Lorenz, Rössler, a Lorenz-like system, and 21 MIT-BIH records, reporting average RMSE 0.0022 on chaotic data and 0.0082 on ECG.

Three methodological caveats are relevant when comparing against this work. First, their prediction horizon is one time step from a 10-step history; on ECG this is a 2 ms forecast. Second, they apply no preprocessing to the ECG, so reported accuracy includes baseline wander (quantified in Section 4.7). Third, no naive baseline is reported; at lag 1 on ECG sampled at 360 Hz, persistence alone attains R^2 in the range 0.98–0.998, which brackets their reported 0.9908.

Fu et al. [2] proposed FGNet, a Fourier-attention encoder with a GRU decoder, evaluated at 1, 4, 7 and 10 steps on Mackey–Glass, Lorenz and sunspot series. Their reported Lorenz R^2 values are higher than those obtained here at matched horizons (Section 4.6). However, their reported Lorenz largest Lyapunov exponent is 0.012285, whereas the accepted value for $(\sigma, \rho, \beta) = (10, 28, 8/3)$ is 0.9056 — a factor of roughly 74, consistent with a per-sample versus per-time-unit unit convention that is not stated. With the correct exponent, their maximum horizon of 10 steps at $dt = 0.01$ corresponds to 0.09 Lyapunov times, i.e. entirely within the short-term-predictable regime.

2.4 Identified Gap

Across the reviewed work, three patterns recur: evaluation at a single short horizon, absence of naive baselines, and horizons that are not expressed in units comparable across systems. No reviewed work combines an NGRC polynomial front-end with a deep TCN encoder–decoder for multi-step waveform forecasting, nor reports a predictability curve in Lyapunov time. That is the gap addressed here.

3 Method

3.1 Architecture

The pipeline is:

Signal $\to$ PSR $\to$ NGRC feature map $\to$ TCN encoder $\to$ interpolate $\to$ TCN decoder $\to$ forecast.

Chaos front-end (zero learnable parameters). The delay embedding is expanded with all unique degree-2 monomials:

$$\Phi(t) = [\underbrace{v_1, \ldots, v_d}_{d \text{ linear}},\ \underbrace{v_1^2, v_1 v_2, \ldots, v_d^2}_{d(d+1)/2 \text{ quadratic}}]. \tag{2}$$

The quadratic terms are motivated by the structure of the Lorenz equations themselves, which contain the products xy and xz; a purely linear map cannot represent these.

Encoder. Four residual TCN blocks, dilations $[1, 2, 4, 8]$, kernel 7, 64 channels, causal.

Decoder. Four non-causal blocks of the same structure. The encoder output is resampled along the time axis by linear interpolation to the target length, followed by a 1×1 convolution to a single channel. This interpolation is the source of the pathology reported in Section 4.3.

3.2 Evaluation Protocol

Every experiment reports at minimum: the model, a zeros/mean baseline, and a persistence baseline. Horizons are converted to Lyapunov time via $\lambda_{\max} t$. On Lorenz, $\lambda_{\max} = 0.9056$ and $dt = 0.01$, giving one Lyapunov time = 110.4 samples.

4 Experimental Trials

4.1 Trial 1 — Initial ECG Forecasting (Failure)

The first experiment forecast 720 samples (2 s) of MIT-BIH ECG from 1080 samples (3 s) of history, using 46 records with an intra-patient 70/10/20 temporal split and per-patient z-scoring fitted on the training portion.

Table 1: Trial 1. Negative R^2 indicates performance worse than predicting the mean.

Model	**MSE**	**RMSE**	**MAE**	R^2
ChaosForecaster (466k params)	1.1475	1.0712	0.6243	−0.0823

Training loss plateaued by epoch 13 and did not improve for a further 30 epochs. The ratio MAE/RMSE = 0.58 (versus ≈ 0.8 for Gaussian errors) indicates heavy-tailed residuals: small errors on the isoelectric baseline and very large errors at QRS complexes. The model had converged to predicting a smooth baseline and omitting every R-peak.

No baseline had been computed, so it was not possible to determine whether the failure was architectural or task-intrinsic.

4.2 Trial 2 — Lorenz Validation

The identical architecture was applied to the univariate $x(t)$ component of the Lorenz system: deterministic, noise-free, with a textbook $\lambda_{\max}$.

PSR parameters converged cleanly ($\tau = 16$, $d = 3$, 9 features), in contrast to the ECG run where FNN saturated at the search ceiling ($d = 10$) for nearly every patient. This confirmed the

AMI/FNN implementation was correct and that ECG saturation reflects residual noise in the data.

Table 2: Trial 2, input length 540 samples.

H	$\lambda_{\max}t$	Chaos R^2	NGRC R^2	Persistence R^2
25	0.23	+0.9602	+0.6932	+0.5345
110	1.00	+0.3633	+0.1565	−0.3092
550	4.98	−0.0135	+0.0270	−1.1075

The architecture worked, but degraded sharply by one Lyapunov time.

4.3 Trial 3 — Input-Length Ablation

A linear autoregressive control on 64 raw lags achieved $R^2 = 0.2565$ at $H = 110$, close to the model's 0.3633, prompting an ablation on input length at fixed $H = 110$.

Table 3: Trial 3. Shorter input is dramatically better — the opposite of the usual expectation that more context helps.

Input length	R^2 **at** $H = 110$
64	0.9590
128	0.9271
200	0.8180
320	0.4056
540	0.3367

Mechanism. The decoder resamples the encoder output from length L to length H by linear interpolation. Output index t is therefore anchored to input index $t \cdot L/H$. With $L = 540$ and $H = 110$, the first predicted sample is constructed from the oldest position in the input window, 5.4 time units (≈ 4.9 Lyapunov times) in the past, where the signal is decorrelated from the present. With $L = 64$ the entire window spans 0.64 time units and remains correlated. The long window was not neutral padding; it actively misaligned the decoder.

A secondary consideration: with dilations $[1, 2, 4, 8]$ and kernel 7, the encoder receptive field is approximately $1 + 2(7 - 1)(1 + 2 + 4 + 8) = 181$ samples, so inputs beyond that length are partially unreachable regardless.

4.4 Trial 4 — Corrected Lorenz Sweep and Full Ablation

With input length set to 64, the sweep was repeated with all arms. Table 4 reports R^2, RMSE, and MAE for each model.

Table 4: Trial 4, input length 64. Full comparison with RMSE and MAE.

		ChaosForecaster			**VanillaTCN**			**Pure NGRC**			**Lin. AR**	**Persist**
H	λt	R^2	RMSE	MAE	R^2	RMSE	MAE	R^2	RMSE	MAE	R^2	R^2
25	0.23	0.995	0.069	0.051	**0.996**	**0.065**	**0.048**	0.693	0.556	0.407	0.916	0.533
110	1.00	**0.956**	**0.210**	**0.130**	0.738	0.512	0.287	0.157	0.919	0.702	0.246	−0.434
550	4.98	**0.075**	**0.963**	**0.766**	0.018	0.992	0.797	0.027	0.987	0.789	0.039	−1.108

Three readings follow directly:

- **At 0.23 Lyapunov times the NGRC front-end contributes nothing.** $R^2 = 0.995$ versus 0.996 is within run-to-run variation (± 0.003 observed across repeated runs). Short-horizon Lorenz-x is close to linearly predictable (0.916 for linear AR). VanillaTCN achieves the lowest RMSE (0.065) and MAE (0.048).

- **At one Lyapunov time the front-end contributes substantially.** $R^2 = 0.956$ versus 0.738 corresponds to an RMSE reduction from 0.512 to 0.210 (59%). This is where the dynamics must actually be propagated forward, and where the explicit quadratic terms match the quadratic structure of the governing equations.

- **At five Lyapunov times all methods collapse.** Values near zero are not meaningfully distinguishable. This horizon was additionally underpowered (27 validation and 54 test windows) and did not converge.

The recursive NGRC ridge model, with only 10 coefficients, reached $R^2 = 0.693$ at $0.23\lambda_{\max}t$. Its underperformance relative to the deep model is consistent with the documented limitation of degree-2 monomials under univariate partial observation.

4.5 Validation Checks

Shuffled-target control. Training the same model on inputs paired with permuted targets yielded $R^2 = -0.0051$ against $+0.9955$ for the real model, confirming no information leakage through window indexing, the PSR offset, or normalisation.

Attractor reconstruction. Stitched predictions embedded at lag $\tau = 16$ reproduce the two-lobe Lorenz structure, confirming the model captured attractor geometry rather than only pointwise amplitude. The predicted embedding is visibly noisier at fine scale than the ground truth.

Lyapunov exponent recovery. A Rosenstein estimator was first calibrated on the true series, recovering $\lambda_{\max} = 0.9105$ against the textbook 0.9056 (0.5% error) with a fit window of [50:200], stable across four (d, τ) settings. Earlier fit windows returned values up to 1.62, an overestimate of 78% — a sensitivity that plausibly explains the divergent exponents reported in the literature. Applied to predictions, the estimator returned 0.6257 against 0.9552 from the truth, a 34.5% underestimate. Two causes apply: squared-error training damps fine-scale divergence, and the stitched trajectory is a concatenation of teacher-forced segments rather than a free-running trajectory, which inflates the initial neighbour distance.

Forecast horizon. Using the σ-crossing criterion (first step at which absolute error exceeds one standard deviation), the model tracked for a mean of 106 steps at $H = 110$, i.e. ≥ 0.96 Lyapunov times. This is a lower bound: 94% of windows never diverged within the available horizon.

4.6 Comparison at FGNet Horizons

Re-running at the horizons used by Fu et al. [2] gives a directly comparable setting (same system, same dt, scale-invariant metric):

Table 5: FGNet achieves higher accuracy at these horizons.

H	$\lambda_{\max}t$	This work	FGNet	Persistence
1	0.01	0.9965	0.999994	0.9970
7	0.06	0.9947	0.999569	0.9421
10	0.09	0.9957	0.998631	0.8923

Two observations. First, at $H = 1$ persistence (0.9970) exceeds this work (0.9965), indicating the regime is close to trivial. Second, the present model's R^2 is essentially flat across $H = 1, 7, 10$ while persistence falls from 0.997 to 0.892; this indicates a fixed error floor near RMSE = 0.06 arising from the decoder rather than from horizon difficulty, and represents an identified implementation gap. All FGNet horizons fall below 0.1 Lyapunov times.

4.7 ECG Baselines and the Baseline-Wander Effect

Persistence baselines were computed on MIT-BIH with input length 360, both with and without the filtering pipeline.

Table 6: Persistence on MIT-BIH. Filtering makes the task measurably harder.

H	seconds	beats	Persist. R^2 (filtered)	Persist. R^2 (raw)
36	0.10	0.12	−0.5398	−0.2023
72	0.20	0.23	−0.8080	−0.4074
180	0.50	0.58	−1.0188	−0.5851
360	1.00	1.17	−0.9993	−0.6186
720	2.00	2.33	−1.0289	−0.7280

Two conclusions follow.

First, the Trial 1 result is re-interpreted: at $H = 720$ the model scored −0.0823 against a persistence baseline of −1.0289. The model comfortably outperformed the standard naive baseline; it failed only against the mean predictor, which on a signal that is near-isoelectric roughly 90% of the time is an unusually strong reference.

Second, removing sub-0.5 Hz drift costs approximately 0.3 in persistence R^2 across all horizons. Baseline wander is electrode and respiration artifact, not cardiac information, yet it is highly autocorrelated and therefore easy to forecast. Published ECG forecasting results obtained without high-pass filtering are inflated by this component.

5 Findings

5.1 Why ECG Forecasting Fails at Multi-Second Horizons

On a z-scored ECG, a QRS complex reaches roughly 3σ over about 30 samples. Under squared error, placing a spike 20 samples from its true location costs $(3 - 0)^2 + (0 - 3)^2 = 18$, whereas predicting a flat line costs $(0 - 3)^2 = 9$. A misplaced spike is penalised twice — once where it is drawn and once where it is missed. Once timing uncertainty exceeds QRS width, the loss-optimal output is no spike at all. The model did not fail to optimise; it optimised correctly under a misspecified objective.

RR-interval variability is 20–50 ms even in normal sinus rhythm and larger in an arrhythmia database. At 360 Hz, 50 ms is 18 samples, already comparable to QRS width by the second beat. The persistence values provide direct evidence. For two uncorrelated samples, $E[(a-b)^2] = 2\sigma^2$, giving $R^2 \approx -1$. Filtered persistence reaches −1.02 by $H = 180$, indicating that ECG samples half a second apart are statistically independent. No architecture can extract information that is not present.

5.2 Chaos Does Not Explain the ECG Failure

Reported ECG largest Lyapunov exponents of order $0.07\,\mathrm{s}^{-1}$ place a 2-second forecast at approximately 0.14 Lyapunov times. The same architecture achieved $R^2 = 0.996$ on Lorenz at 0.23 Lyapunov times — a longer horizon in chaos units — while ECG returned −0.0823 at a shorter one. Sensitive dependence on initial conditions therefore does not account for the discrepancy. The binding constraint is stochastic beat timing, which is not a chaotic phenomenon.

6 Limitations and Next Steps

1. **Coarse horizon sweep.** Only three horizons were run; the crossing point of $R^2 = 0$ cannot be localised. Intermediate horizons (55, 220) and longer ones (1100, 2200) are required.
2. **Underpowered long horizon.** $H = 550$ had 27 validation and 54 test windows and did not converge. That row should not be treated as a measurement.
3. **Fixed error floor.** R^2 is flat across $H = 1, 7, 10$, indicating a decoder-side resolution limit independent of horizon.
4. **Interpolation decoder.** The identified anchoring pathology was mitigated by shortening the input, not removed. A learned projection from the encoder's final state would address it at the source.
5. **Non-causal filtering before splitting.** `filtfilt` and a globally estimated SWT threshold allow mild information flow across split boundaries.
6. **Univariate NGRC.** Degree-2 monomials are known to be insufficient under partial observation; a Chebyshev basis is the documented remedy.
7. **ECG horizon not yet swept.** The predictability curve for ECG at $H \in \{36, 72, 180, 360\}$ remains to be measured.

References

[1] H. V. Dudukcu, M. Taskiran, Z. G. C. Taskiran, and T. Yildirim, "Temporal Convolutional Networks with RNN approach for chaotic time series prediction," *Applied Soft Computing*, vol. 133, p. 109945, 2023.

[2] K. Fu, H. Li, and X. Shi, "An encoder–decoder architecture with Fourier attention for chaotic time series multi-step prediction," *Applied Soft Computing*, vol. 156, p. 111409, 2024.

[3] D. J. Gauthier, E. Bollt, A. Griffith, and W. A. S. Barbosa, "Next generation reservoir computing," *Nature Communications*, vol. 12, p. 5564, 2021.

[4] H. Khoshroo and Y. Shekofteh, "Chaotic time series prediction using combination of spatial and temporal information by deep learning approaches," *Nonlinear Dynamics*, vol. 113, pp. 17365–17383, 2025.

[5] H. N. Phien, T. T. Dang, and N. Q. K. Le, "Strategies of multi-step-ahead forecasting for chaotic time series using autoencoder and LSTM neural networks: A comparative study," in *Proc. IPMV*, 2023.

[6] B. Ramadevi and K. Bingi, "Chaotic time series forecasting approaches using machine learning techniques: A review," *Symmetry*, vol. 14, no. 5, p. 955, 2022.

[7] S. Shahi, F. H. Fenton, and E. M. Cherry, "Prediction of chaotic time series using recurrent neural networks and reservoir computing techniques: A comparative study," *Machine Learning with Applications*, vol. 8, p. 100300, 2022.

[8] X. Wang, M. Yang, Y. Zeng, Z. Lin, and H. H. C. Iu, "An effective ECG signal classification method based on a minimalistic memristive reservoir computing system," *Cognitive Neurodynamics*, vol. 19, p. 97, 2025.